Lab 06 - Aggregation

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Idea

The goal is to make a swarm of robots aggregate into a single cluster using only local interactions, without any centralised control.
Each robot runs a stochastic two-state FSM (moving/stopped). Transition probabilities are modulated by the number of stopped neighbours perceived via the range-and-bearing (RAB) sensor: the more stopped robots nearby, the more likely a moving robot stops, and the less likely a stopped robot resumes walking. Positive feedback promotes cluster growth, while negative feedback prevents small unstable aggregates from persisting. Together they drive the swarm toward the formation of one dominant cluster.

Controller architecture

The controller (`aggregation.lua`) implements the following components:

- **State machine**: two states, `STATE_MOVING` and `STATE_STOPPED`.
- **Neighbour counting (`CountRAB`)**: counts robots within `MAXRANGE = 30 cm` that broadcast data byte `1` (stopped). Moving robots broadcast `0`.
- **Stopping probability**: `Ps = min(PSmax, S + alpha·N)` increases with `N`, base value `S = 0.01`, `alpha = 0.4`, cap `PSmax = 0.99`.
- **Walking probability**: `Pw = max(PWmin, W - beta·N)` decreases with `N`, base value `W = 0.1`, `beta = 0.05`, floor `PWmin = 0.005`.
- **Obstacle avoidance**: active only when moving. It detects the strongest front proximity reading (`|angle| < π/2`, threshold `0.1`) and turns away from it. If no obstacle, the robot moves straight at `SPEED = 10`.
- **LED feedback**: green when moving, red when stopped.

Each step, `N` is sampled, `Ps` and `Pw` are recomputed, and a uniform random draw decides the state transition.

Expected behavior

- Isolated moving robots rarely stop (low `Ps` with `N = 0`) and keep exploring.
- A robot near a stopped cluster perceives high `N`, so `Ps` rises sharply and it is likely to stop and join the cluster.
- Stopped robots in small groups (low `N`) have a relatively high `Pw` and may resume walking, dissolving small clusters. The algorithm naturally favours larger, stable aggregates.
- Over time the swarm converges to one dominant cluster through this self-reinforcing local feedback.